Project Title: Predictive Pulse: Harnessing Machine Learning for Blood Pressure Analysis

Project Design Phase-I - Solution Fit Template

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Define CS, fit into CC

1. CUSTOMER SEGMENT(S) CS

Who is your customer?
i.e. working parents of 0-5 y.o. kids

Primary users of the system include:

- Healthcare professionals such as doctors, nurses, and medical staff
- Hospitals and clinics are conducting routine patient monitoring
- Individuals with hypertension risk factors
- Preventive healthcare organizations conducting health screenings

6. CUSTOMER CONSTRAINTS CC

What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices.

Customers may face the following constraints:

- Limited access to advanced AI tools
- Lack of technical expertise in machine learning
- Time constraints in busy hospital environments
- Data availability and integration issues
- Need for simple and easy-to-use interfaces

5. AVAILABLE SOLUTIONS AS

Which solutions are available to the customers when they face the problem
or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking

Existing solutions used by healthcare professionals include:

- Manual blood pressure monitoring
- Traditional clinical diagnosis methods
- Basic hospital record systems
- Wearable health monitoring devices

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMS J&P

Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides.

The customers need to:

- Quickly analyze patient blood pressure data
- Detect early stages of hypertension
- Support clinical decision-making
- Identify patients requiring immediate intervention
- Reduce time taken for manual health assessment

9. PROBLEM ROOT CAUSE RC

What is the real reason that this problem exists?
What is the back story behind the need to do this job?
i.e. customers have to do it because of the change in regulations.

The root causes of the problem include:

- Growing prevalence of hypertension worldwide
- Lack of early detection systems
- Dependence on manual analysis
- Limited predictive analytics in traditional healthcare systems
- Increasing workload on healthcare professionals

7. BEHAVIOUR BE

What does your customer do to address the problem and get the job done?
i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace)

Typical behavior of users when dealing with hypertension analysis:

- Doctors examine patient medical history
- Measure blood pressure manually
- Analyze symptoms and lifestyle factors
- Use clinical judgment for diagnosis
- Maintain patient records in hospital systems

Healthcare professionals increasingly explore digital tools and predictive analytics for improved diagnosis.

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

3. TRIGGERS TR

What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.

Events that encourage customers to use the system include rising hypertension cases, the need for early disease detection, faster patient assessment in hospitals, increased adoption of AI in healthcare, and preventive health screening programs.

4. EMOTIONS: BEFORE / AFTER EM

How do customers feel when they face a problem or a job and afterwards?
i.e. lost, insecure > confident, in control - use it in your communication strategy & design.

Before using the system:

- Doctors may feel uncertain or pressured due to large patient loads.
- Patients may feel anxious or unaware of their health risks.

After using the system:

- Doctors feel more confident in diagnosis and treatment decisions.
- Patients feel informed and reassured about their health condition.

10. YOUR SOLUTION SL

If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality.
If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.

The proposed solution is a machine learning-based hypertension prediction system.

Key features include:

- Uses patient health data and clinical parameters
- Applies machine learning algorithms to predict hypertension risk
- Provides quick and accurate predictions
- Deployed as a Flask-based web application
- Helps doctors make data-driven medical decisions

The system improves early detection, preventive healthcare, and clinical efficiency.

8. CHANNELS of BEHAVIOUR CH

8.1 ONLINE

What kind of actions do customers take online? Extract online channels from #7

8.2 OFFLINE

What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.

Online

- Users may interact with the solution through:
- Hospital digital health systems
- Web-based medical tools
- Online health monitoring platforms
- Medical data analysis dashboards

Offline

- Offline interactions include:
- Hospital consultations
- Routine health check-ups
- Community health screening programs
- Clinical diagnosis during patient visits

Identify strong TR & EM